NomNomSafe

Final Presentation

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ASE 485 Capstone

What Problem Did I Solve?

- Restaurants need frequent menu updates, but sources are messy (PDF, DOCX, CSV, URL text).
- Manual entry is slow and error-prone.
- Allergen mistakes are safety-critical.
- The core problem: turn messy menu data into trustworthy, reviewable allergen-aware records.

Why This Matters

- A wrong allergen signal can harm real people.
- Businesses need confidence and accountability in their menu data.
- "Unknown" should not be treated as "safe."
- The project goal was practical safety, not just feature completion.

My Semester Approach

- Two-track plan:
  - **Track A (software):** build and validate business workflows.
  - **Track B (research):** build evidence-based allergen reasoning.
- Use AI as often as it made sense, and dynamically and critically assess how AI is used.
- Human confirmation remained mandatory for safety-relevant fields.

Progress Since S1P

- Sprint 1 delivered a stable MVP foundation:
  - archived non-MVP features,
  - refactored front-end structure,
  - improved search/filter/sort,
  - improved onboarding.
- Sprint 2 expanded capability:
  - AI-assisted menu import/review/save,
  - stronger reliability/security controls,
  - ontology research completed (Phases 0-2).

Project Results and Validation

- End-to-end flow works: import -> review -> confirm -> save.
- Multiple import paths supported (file and URL with fallback behavior).
- Reliability improvements added: quality gates, timeout/retry, rate limiting, clearer error handling.
- Validation completed via weekly checkpoints, regression checks, and demo readiness checklist.

The System Prompt

const SYSTEM_PROMPT = `You are a menu data extraction assistant. Your job is to parse raw text from restaurant menus and return structured JSON.

Return ONLY a valid JSON array. No markdown, no backticks, no explanation.
Each item must have exactly these fields:
{
"name": "string (required — item name)",
"description": "string (use empty string if not found)",
"ingredients": ["array of strings — each ingredient as a separate string. Use empty array [] if not found."],
"price": "number or null (numeric value only, no currency symbols)",
"category": "string (infer from context, e.g. 'Appetizers', 'Entrees', 'Desserts', 'Drinks'. Use 'Uncategorized' if unknown.)",

CRITICAL RULES:

- ingredients MUST be a JSON array of strings, never a single string.
- possible_allergens should be conservative but useful:
  i. Include allergens explicitly stated in the source text (e.g., "contains milk").
  ii. Also include allergens inferred from clear ingredient keywords in the same item text.
- Use only this controlled ingredient-to-allergen mapping:
  - milk, buttermilk, butter, cream, cheese, yogurt, whey, casein, ghee -> milk
  - wheat, flour, bread, pasta, noodle, semolina, durum, breadcrumbs -> gluten
  - egg, eggs, mayonnaise, aioli -> egg
  - peanut, peanuts, peanut butter -> peanuts
  - almond, walnut, pecan, pistachio, cashew, hazelnut, macadamia -> tree nuts
  - soy, soya, soybean, tofu, edamame, miso, tempeh -> soy
  - fish, salmon, tuna, cod, anchovy, sardine -> fish
  - shrimp, prawn, crab, lobster, crayfish, clam, mussel, oyster, scallop -> shellfish
  - sesame, tahini -> sesame
  - mustard -> mustard
  - celery -> celery
  - lupin -> lupin
  - sulfite, sulphite, sulfites, sulphites -> sulfites

How the System Double-Checks AI Imports

Safe files, safe links, steady behavior

- You can import from **supported file types** (such as PDF, Word, or spreadsheet) up to a **reasonable file size**—not unlimited uploads.
- **Web links** are opened carefully: blocked risky hosts, timeouts, and rate limits help prevent abuse and long hangs.
- If a file or page has **no readable text** (for example a blank export or a scan with no text), you get a **clear message** instead of a mystery result.

Cleaning up each suggested dish

- Every row must have a **dish name**; rows without a name are dropped.
- **Same name, same section twice?** Flagged as a duplicate so your list does not fill with clones.
- Long text is **trimmed** to sensible limits; messy ingredient lines are split into a **simple list** where possible.
- **Prices** like “$12.99” are turned into numbers when the system can read them; if it cannot, the price stays **empty**—the system does not guess a number.
- **Allergen hints** are lined up with **standard names** when they clearly match (for example “eggs” becomes egg).

When to pause and review

- If **nothing usable** is left after cleanup, import **stops** and suggests another source or manual entry—**no empty shell pretending to be a menu**.

Demo (Live or Video)

Demonstration Link

- Video: https://youtu.be/MfqT1rB7Udw

What the demo shows

- Successful import from clean source.
- Human-confirmed allergen decisions before save.

Dashboard › **Setup**

Step 1/2 – Business info

Basic Business Information

Find your business

Start by searching for your business or address.

Four Mile Pig, Alexandria Pike, Alexandria, KY, USA

Powered by Google

We've filled in the details below. Review and edit if needed.

Business name & website

Four Mile Pig

http://www.fourmilepig.com/

Business address

All address fields are required.

7962 Alexandria Pike

Alexandria | KY | 41001

Tips & notes

Back | Continue

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Dashboard › **Menu Items**

Import

+ Add Item

Your Menu

Search by name, description, ingredients, or allergens...

Filters

All | Appetizers | Entrees | Sides | Desserts | Drinks

No items in this menu yet.

Add your first item

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Dashboard › Menu Items › **Import**

Import menu

Add many items at once. After parsing, review every line before saving. Target menu: Your Menu

File | **URL** | Manual

Enter a public page URL. Private networks are blocked.

https://www.fourmilepig.com/menu?menu=menu-3

Import from URL

Type URL here

Click this button

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Dashboard › Menu Items › **Import**

Import menu

Add many items at once. After parsing, review every line before saving. Target menu: Your Menu

Parsing items...

File | **URL** | Manual

Enter a public page URL. Private networks are blocked.

https://www.fourmilepig.com/menu?menu=menu-3

Loading...

This can take up to 20 seconds for larger files.

Back

Review imported items

Target: Your Menu · Unselect any row you do not want. Parser suggested allergens are not saved until you confirm the checkbox and select database allergens to store.

36 item(s) have parser allergen suggestions.

Match suggestions to allergen checkboxes (all items)

This auto-populates the entire list of menu items with suggestions from the AI

Include in save OK

If an item snuck through or you otherwise dont want it included, uncheck this

NAME

Hickory Smoked Pulled Pork

PRICE

7.5

CATEGORY

Sandwiches & Burgers

DESCRIPTION

Served on toasted brioche bun with pickles and sauce on the side.

INGREDIENTS (COMMA-SEPARATED)

smoked pulled pork, toasted brioche bun, pickles, sauce

Allergens

Parser suggestions are hints only. Nothing is stored until you pick database allergens and confirm below. Use "Match suggestions" to pre-fill checkboxes from the parser text, then review and adjust.

PARSER SUGGESTIONS (UNTRUSTED)

gluten, milk, egg

Match suggestions to checkboxes

Match suggestions from the AI parser at the item level

Gluten	Dairy	Egg	Soy	Peanut
Tree nut	Fish	Shellfish	Sesame	

No allergens selected

Include in save OK

Back

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☑ Include in save OK

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PARSER SUGGESTIONS (UNTRUSTED)

gluten, milk, egg

Match suggestions to checkboxes

Selected:

Gluten × Dairy × Egg ×

☑ Gluten	☑ Dairy	☑ Egg	☐ Soy	☐ Peanut
☐ Tree nut	☐ Fish	☐ Shellfish	☐ Sesame	

☑ Include in save OK

Gluten | Dairy | Egg | Soy | Peanut
Tree nut | Fish | Shellfish | Sesame

Include in save OK

NAME

Wrapped Double Fudge Brownie

PRICE

2

CATEGORY

Dessert

DESCRIPTION

INGREDIENTS (COMMA-SEPARATED)

double fudge brownie

Allergens

Parser suggestions are hints only. Nothing is stored until you pick database allergens and confirm below. Use "Match suggestions" to pre-fill checkboxes from the parser text, then review and adjust.

PARSER SUGGESTIONS (UNTRUSTED)

gluten, milk, egg

Match suggestions to checkboxes

Selected:

Gluten × Dairy × Egg ×

Gluten | Dairy | Egg | Soy | Peanut
Tree nut | Fish | Shellfish | Sesame

I have reviewed allergen information for selected items (required before save when parser suggested allergens)

Save selected to menu

Have to check this box before saving menu items to a menu

Green Beans $7.00

By The Pound

No allergens

Entree

Grilled Cheese & Bacon $6.50

Three cheeses & bacon grilled on sourdough

Gluten Dairy

Entree

Grilled cheese & fries $4.00

Gluten Dairy

Entree

Hickory Smoked Pulled Pork $7.50

Served on toasted brioche bun with pickles and sauce on the side.

Gluten Dairy Egg

Entree

Hickory Smoked Pulled Pork $16.00

By The Pound

No allergens

Entree

Kicking Chicken Sandwich $8.95

Smoked chicken breast with spicy BBQ, melted swiss cheese, pickles & onions on toasted brioche bun

Gluten Dairy Egg

Entree

Loaded Fries $11.95

Smoked pork, basket of fries, shredded cheese and topped with BBQ sauce!

Dairy

Entree

Learning with AI - Topic 1

AI-Assisted Structured Data Extraction

- I learned AI can convert messy menu text into usable structured records.
- I also learned model output must be treated as untrusted input.
- Best result came from combining:
  - AI extraction for speed,
  - validation rules for quality,
  - human review for safety-critical decisions.

Learning with AI - Topic 2

Ontology Research Deliverables (Why They Matter)

The real question

- "Can this dish be called safe for an allergen?"

Why this needed research

- Menu wording is often incomplete.
- Staff language can be inconsistent.
- AI can sound confident even when evidence is missing.

What the deliverables do

- Turn vague language into a clear evidence process.
- Separate "what we know" from "what we are guessing."
- Require proof before any safety claim is allowed.

Learning with AI - Topic 2

Documents Explained

Deliverable	Plain-language outcome
R3.1 - Failure Taxonomy	Identified common unsafe reasoning patterns (for example, treating "not listed" as "safe").
R3.2 - Evidence Requirements	Defined the minimum evidence needed before a conclusion can be trusted.
R3.3 - Core Exposure Model	Built a simple state model (including **Unknown**) with a strict rule for when "safe" is allowed.
R3.4 - Scenario Test Suite	Validated the model on 18 realistic food-service scenarios to confirm behavior under ambiguity.
R3.5 - Communication Layer	Mapped real menu/staff/disclaimer language into structured evidence without changing the core safety rule.

Bottom line

- The research produced a reusable safety framework that supports AI assistance while protecting against false confidence.

Issues I Faced and How I Solved Them

- **Scope pressure:** solved by archiving non-MVP work and keeping a clear priority order.
- **Safety ambiguity:** solved by formal reasoning rules plus mandatory human confirmation.
- **Late-stage risk:** solved by a Week 14 scope freeze and explicit known limitations.

Public Artifacts and Evidence

- Canvas page contains work presented at NKU Celebration of Research Thursday April 23.
- Presentation led by project collaborator Anna Dinius

Closing

Q and A